UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Classes 003–004 · Total derivative, chain rule and implicit functions · compiled 14 October 2026

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1 Short notes

1 · Composite functions and the one-variable chain rule

A composite function is a function of a function, e.g. $y = e^{\sin x^2}$. Its derivative is found from the outside in and the factors are multiplied: $e^{\sin x^2} \cdot \cos x^2 \cdot 2x$. For several variables the same idea gives the total derivative and the chain rule.

2 · The total derivative rule

If $z = f(x, y)$ and $x = x(t)$, $y = y(t)$, then z is a function of t alone and its total derivative is $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. A tree diagram with one branch per intermediate variable gives the terms. The answer is written in terms of t or evaluated at the given t.

3 · The total differential: approximation and sensitivity

The total differential $dz = f_x\,dx + f_y\,dy$ (or $\sum f_{x_i} dx_i$) gives the approximate change in z for small changes in the variables. It approximates nearby values, $f(x_0 + dx, y_0 + dy) \approx f + f_x\,dx + f_y\,dy$. It also measures sensitivity: the larger coefficient marks the variable that matters most. It is used in substitution and error estimates.

4 · Rates of change problems

For a rate problem, write the quantity as a function of the changing variables, take d/dt of it with the total derivative rule, and substitute the given values and rates (decreasing means negative). If something stays constant its derivative is 0; a constraint between variables gives a missing rate by implicit differentiation.

5 · The chain rule for composite functions

If $z = f(x, y)$ with $x = g(u, v)$, $y = h(u, v)$, then z is a composite function of u, v and $z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$ (one term per path in the tree diagram). With three intermediate variables there are three terms. If f is unknown, keep f_x, f_y as symbols. Write the final answer in the new variables. Standard results: $f(x - y, y - z, z - x)$ gives $u_x + u_y + u_z = 0$; in polar form $u_x^2 + u_y^2 = u_r^2 + u_\theta^2 / r^2$.

6 · Implicit functions of two variables

An implicit function is given by a relation $f(x, y) = c$, a level curve. Along it $df = f_x\,dx + f_y\,dy = 0$, so $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$ ($f_y \neq 0$). Direct differentiation of both sides gives the same result. The slope at (x_0, y_0) gives the tangent $y - y_0 = m(x - x_0)$; e.g. $xe^y + ye^x = 0$ has tangent $x + y = 0$ at the origin.

7 · Implicit functions of three variables

For $f(x, y, z) = 0$, z is a function of x and y: $\dfrac{\partial z}{\partial x} = -\dfrac{f_x}{f_z}$ (hold y), $\dfrac{\partial z}{\partial y} = -\dfrac{f_y}{f_z}$, and $\dfrac{\partial y}{\partial x} = -\dfrac{f_x}{f_y}$ (hold z), provided the denominators are non-zero. Check that the point lies on the surface. For second derivatives, remember that z still depends on x, y.

2 Keyword index

33 terms; the number in brackets is the topic.

Approximation (3). Estimating f near a known point with $f + f_x\,dx + f_y\,dy$.

Branch diagram (6). Diamond diagram for dz/dx when $y = h(x)$: $f_x \cdot 1 + f_y\,y'$.

Chain (of a problem) (5). What depends on what; decides the formula.

Chain rule (one variable) (1). Differentiate outside-in and multiply.

Chain rule (several variables) (5). $z_u = z_x x_u + z_y y_u$.

Composite function (1). Function of functions, e.g. $f(x(u,v), y(u,v))$.

Constraint (4). Relation between variables used to find a missing rate.

Cylinder (6). $x^2 + y^2 = c$; its layers are level surfaces.

Decreasing rate (4). A negative derivative with respect to time.

Differential (3). Another name for the total derivative dz.

Equipotential surface (6). Level surface on which the potential is constant.

Error estimate (3). Use of the differential, covered at the end of the unit.

Explicit function (6). $y = g(x)$ given by a formula.

Hyperbolic paraboloid (4). Saddle surface $x^2/16 - y^2/9 = z$ (the slide says paraboloid).

Identity for any f (5). A result like $u_x + u_y + u_z = 0$ that holds whatever the outer function f is.

Implicit differentiation (6). Finding dy/dx from $f(x,y) = 0$: $-f_x/f_y$.

Implicit function (6). Function given by a relation $f(x,y) = c$.

Integration by substitution (3). Uses the differential: $t = x^2$, $dt = 2x\,dx$.

Intermediate variable (2). Variable between z and the final variables.

Isotherm (6). Level surface of constant temperature.

Level curve (6). $f(x,y) = c$; its slope is $-f_x/f_y$.

Level surface (7). $f(x,y,z) = c$, or $z = f(x,y) = c$.

Parameter (2). The single variable t in the total derivative.

Polar coordinates (5). $x = r\cos\theta$, $y = r\sin\theta$.

Rate of change (4). Derivative with respect to time.

Related rates (4). Rates linked through a formula or a constraint.

Sensitivity (3). Which variable's change affects the output most (largest coefficient in dz).

Sphere (7). $x^2 + y^2 + z^2 = c$; nested level surfaces.

Tangent line (6). $y - y_0 = m(x - x_0)$ with $m = dy/dx$.

Total derivative (2). $dz/dt = z_x\,x' + z_y\,y'$.

Total differential (3). $dz = f_x\,dx + f_y\,dy$.

Tree diagram (2). Root z, branches labelled with derivatives; multiply along, add across.

Vertical branch (6). $dx/dx = 1$ in the branch diagram.

3 Rapid recall

1	$\dfrac{d}{dx} e^{\sin x^2}$?	$2x\, e^{\sin x^2} \cos x^2$.
2	Total derivative of $z = f(x,y)$, $x(t)$, $y(t)$?	$dz/dt = z_x\, dx/dt + z_y\, dy/dt$.
3	Ordinary or partial: z of x, y, both of t?	Ordinary dz/dt.
4	Ordinary or partial: z of x, y, both of u, v?	Partial $\partial z/\partial u$, $\partial z/\partial v$.
5	Chain rule for $\partial z/\partial u$?	$z_x x_u + z_y y_u$.
6	Total differential of $z = f(x,y)$?	$dz = f_x\, dx + f_y\, dy$.
7	Total differential of $u = x/y$?	$(y\, dx - x\, dy)/y^2$.
8	Approximate $f(2.1, -3.03)$ given $f = 6$, $f_x = 1.3$, $f_y = -0.6$ at $(2,-3)$?	6.148.
9	Tank $r = 2$, $h = 10$: more sensitive to r or h?	r: $dV = 40\pi\, dr + 4\pi\, dh$.
10	"Area remains constant" means?	$dA/dt = 0$.
11	A side decreasing at 2 cm/s: its rate?	-2 cm/s.
12	dz/dt for $z = x^2y^3$, $x = t^2$, $y = 2t^3$?	$104t^{12}$ (the slide's $60t^{12}$ is wrong).
13	Cone with $r = h$, $dV/dt = 9$, $h = 6$: dh/dt?	$1/(4\pi) \approx 0.0796$ cm/s.
14	dy/dx for $f(x,y) = c$?	$-f_x/f_y$, if $f_y \neq 0$.
15	dy/dx for $x^2 + xy + y^2 = 7$?	$-(2x+y)/(x+2y)$.
16	Tangent to $xe^y + ye^x = 0$ at the origin?	$x + y = 0$.
17	$\partial z/\partial x$ from $f(x,y,z) = 0$?	$-f_x/f_z$ (hold y).
18	$\partial y/\partial x$ from $f(x,y,z) = 0$?	$-f_x/f_y$ (hold z).
19	$z^3 + xy - y^2 z = 6$ at $(0,1,2)$: z_x?	$-1/11$.
20	$u = f(x-y, y-z, z-x)$: $u_x + u_y + u_z$?	0.
21	$u = f(x/y, y/z, z/x)$: $xu_x + yu_y + zu_z$?	0.
22	$u = f(r)$, $r = \sqrt{x^2+y^2+z^2}$: u_x?	$f'(r)\, x/r$.
23	$u = \tan^{-1}(y/x)$ in polar form: u_θ?	1.
24	$u = e^{x^2+y^2}$ in polar form: u_θ?	0.
25	$u_x^2 + u_y^2$ in polar form?	$u_r^2 + u_\theta^2/r^2$.

Mnemonic (study aid). *"Multiply down, add across."* In a tree diagram, multiply the derivatives along each path from the top to the variable you want, then add the paths.

Mnemonic (study aid). *"Implicit: minus, top over the one you want."* $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$: the variable you differentiate by (x) on top, the one you solve for (y) below.

4 Last 10 minutes before the exam

Top facts

1. Total derivative: $z = f(x, y)$, $x(t)$, $y(t)$ gives $dz/dt = z_x\,dx/dt + z_y\,dy/dt$ (ordinary d).
2. Chain rule: $x(u, v)$, $y(u, v)$ gives $z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$ (partial ∂).
3. Draw the tree diagram and know the chain before writing any formula.
4. Total differential: $dz = f_x\,dx + f_y\,dy$; it approximates changes and nearby values.
5. Rates: decreasing means negative; constant means derivative 0; constraints give missing rates.
6. Implicit, two variables: $dy/dx = -f_x/f_y$ (slide 004·4 has it upside down).
7. Implicit, three variables: $z_x = -f_x/f_z$, $z_y = -f_y/f_z$, $y_x = -f_x/f_y$; check that the point is on the surface.
8. Unknown f: keep f_x, f_y as symbols; eliminate x, y from final answers in r, θ or s, t.
9. Exam favourites: $u_x + u_y + u_z = 0$ for $f(x - y, y - z, z - x)$; $z_x^2 + z_y^2$ identities under $u = x^2 - y^2$, $v = 2xy$ or $x = e^u \sin v$.
10. Slide errors to avoid: $dz/dt = 104t^{12}$ (not $60t^{12}$), $d\theta/dt = -0.0481$ (not 0.04831), $v_y = -3v_r + 3v_s$.

Top formulas

E2.1 · Total derivative rule ESA ★★

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

gives	Rate of change of z with respect to the single variable t.
applies	$z = f(x, y)$, $x = x(t)$, $y = y(t)$.
note	Ordinary d on z, x, y with respect to t; partial ∂ only on z with respect to x, y.

S003·5 · T003a 00:05:54 · T003b 00:01:10

E3.1 · Total differential ESA ★★

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy$$

gives	Approximate change in z for small changes dx, dy.
applies	$z = f(x, y)$ differentiable.
note	n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E5.1 · Chain rule, two intermediate variables ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

GIVES Partial derivatives of a composite function.

APPLIES $z = f(x, y)$, $x = g(u, v)$, $y = h(u, v)$.

NOTE All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E6.1 · Implicit differentiation, two variables ESA ★★

$$f(x, y) = c \implies \frac{dy}{dx} = -\frac{f_x}{f_y}, \qquad f_y \neq 0$$

GIVES Slope of the level curve at a point.

APPLIES $f_y \neq 0$ at the point.

NOTE From $df = f_x\,dx + f_y\,dy = 0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E7.1 · Implicit differentiation, three variables ESA ★★

$$f(x, y, z) = 0: \quad \frac{\partial z}{\partial x} = -\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y} = -\frac{f_y}{f_z}\ (f_z \neq 0); \qquad \frac{\partial y}{\partial x} = -\frac{f_x}{f_y}\ (f_y \neq 0)$$

GIVES Partial derivatives of the implicitly defined variable.

APPLIES Hold y constant for z_x; hold z constant for y_x.

NOTE Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

E5.4 · Polar coordinates: derivatives you reuse ESA ★★

$$x = r\cos\theta,\ y = r\sin\theta: \qquad x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta$$
$$r = \sqrt{x^2 + y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: \quad r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r, \theta)$.

S004·13 · S004·15 · S004·17